Human Review: rank-one map $\ell^1 \to \ell^\infty$

Original automatic proof: `solution_packet.pdf`
Checked date: 14 June 2026
Status: Not independently checked in detail by the human reviewer; excluded from the survey.
Verdict: Excluded from the main survey.

Human Comments

The packet claims to identify a local error in the source paper. The paper states that the rank-one operator

$$T : \ell^1 \to \ell^\infty, \qquad T(x) = \left(\sum_{k=1}^{\infty} x_k\right) \mathbf{1}, \qquad \mathbf{1} = (1, 1, \ldots),$$

is not disjoint weak-star Dunford-Pettis. The automatic packet argues that this assertion is false.

Indeed, write

$$\sigma(x) = \sum_{k=1}^{\infty} x_k.$$

Then

$$T(x) = \sigma(x)\mathbf{1}.$$

If (x_n) is a norm-bounded disjoint sequence in ℓ^1, then $(\sigma(x_n))$ is bounded. If

$$f_n \xrightarrow{w^*} 0 \quad \text{in } (\ell^\infty)',$$

then, since $\mathbf{1} \in \ell^\infty$,

$$f_n(\mathbf{1}) \longrightarrow 0.$$

Therefore

$$f_n(Tx_n) = f_n(\sigma(x_n)\mathbf{1}) = \sigma(x_n) f_n(\mathbf{1}) \longrightarrow 0.$$

This would be exactly the definition of a disjoint weak-star Dunford-Pettis operator, and hence would refute the displayed assertion in the source paper.

The alleged mistake in the source paper appears to come from an incorrect handling of weak-star null sequences in $(\ell^\infty)'$. In particular, one cannot generally represent arbitrary elements of $(\ell^\infty)'$ by countable coordinate sums, and any weak-star null sequence in $(\ell^\infty)'$ must vanish on the constant vector $\mathbf{1}$.

Short Version

The automatic packet gives a short argument that the rank-one map factors through the constant vector $\mathbf{1}$, and that weak-star null functionals on $(\ell^\infty)'$ vanish on $\mathbf{1}$. If correct, this would show that the operator is disjoint weak-star Dunford-Pettis, contrary to the assertion in the source paper.

Literature and Follow-Up

This should be treated, at most, as a possible local correction to a printed example, not as a substantive new mathematical discovery. The surrounding paper also shows some signs of technical unreliability, so this packet is best understood as an example-checking case rather than as progress on an open problem.

Review Outcome

Exclude this packet from the main survey of successful mathematical discoveries. It may be retained as a cautionary example of a possible local paper-error detection, but the argument has not been independently checked in detail by the human reviewer, and in any case the mathematical contribution would be too small and too local to count as a main survey example.